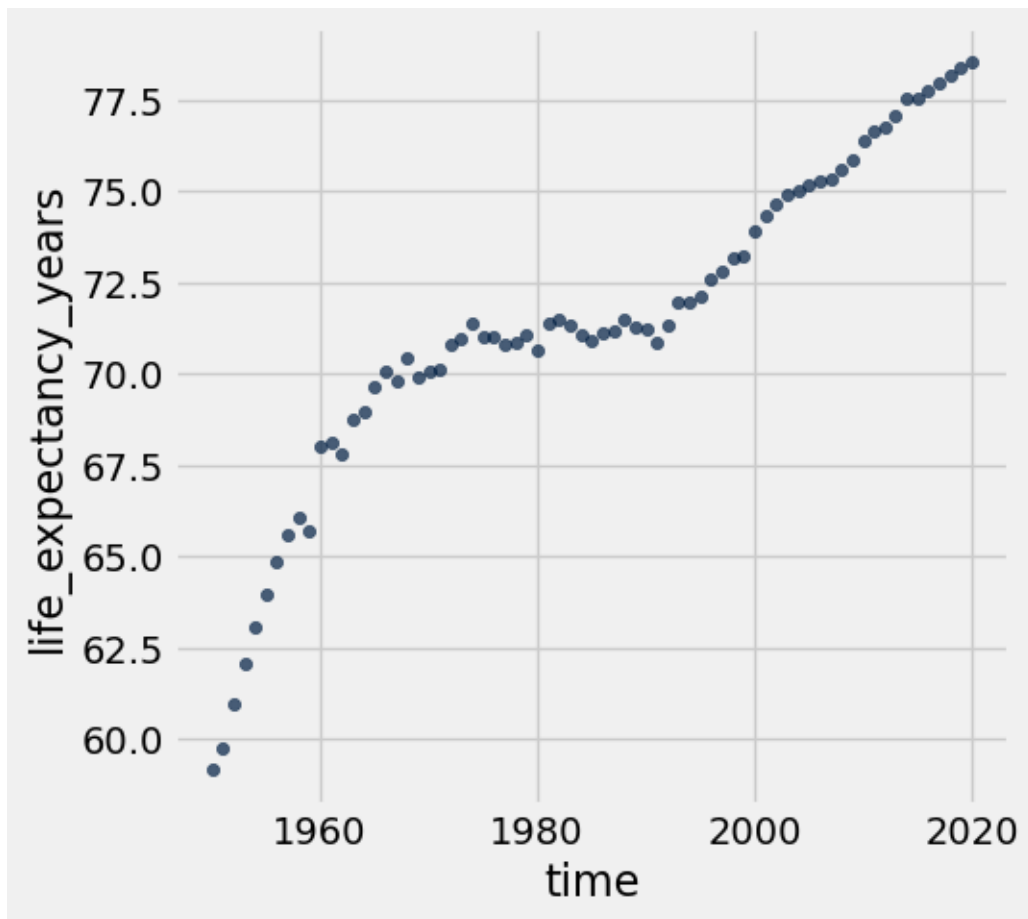


Question 3. Is population growing more slowly perhaps because people aren't living as long? Use the `life_expectancy` table to draw a line graph with the years 1950 and later on the horizontal axis that shows how the *life expectancy at birth* has changed in Poland.

Hint: Make sure you filter the table properly; otherwise, the graph may look funky!

In [83]: *# Fill in code here*

```
life_expectancy.where('geo', are.containing('pol')).where('time', are.above_or_equal_to(1950))
```



Question 4. Assuming everything else stays the same, do the trends in life expectancy in the graph above directly explain why the population growth rate decreased since 1950 in Poland? Why or why not?

The general trend from the life expectancy graph is that the life expectancy of people in Poland increased between 1950 and 2020. But, this has no relation or association to the number of new births in Poland during this period. So, this trend does not directly explain why the population growth rate in Poland has decreased since 1950.

Question 6. Assuming everything else is constant, do the trends in fertility in the graph above help directly explain why the population growth rate decreased from 1950 to 2020 in Poland? Why or why not?

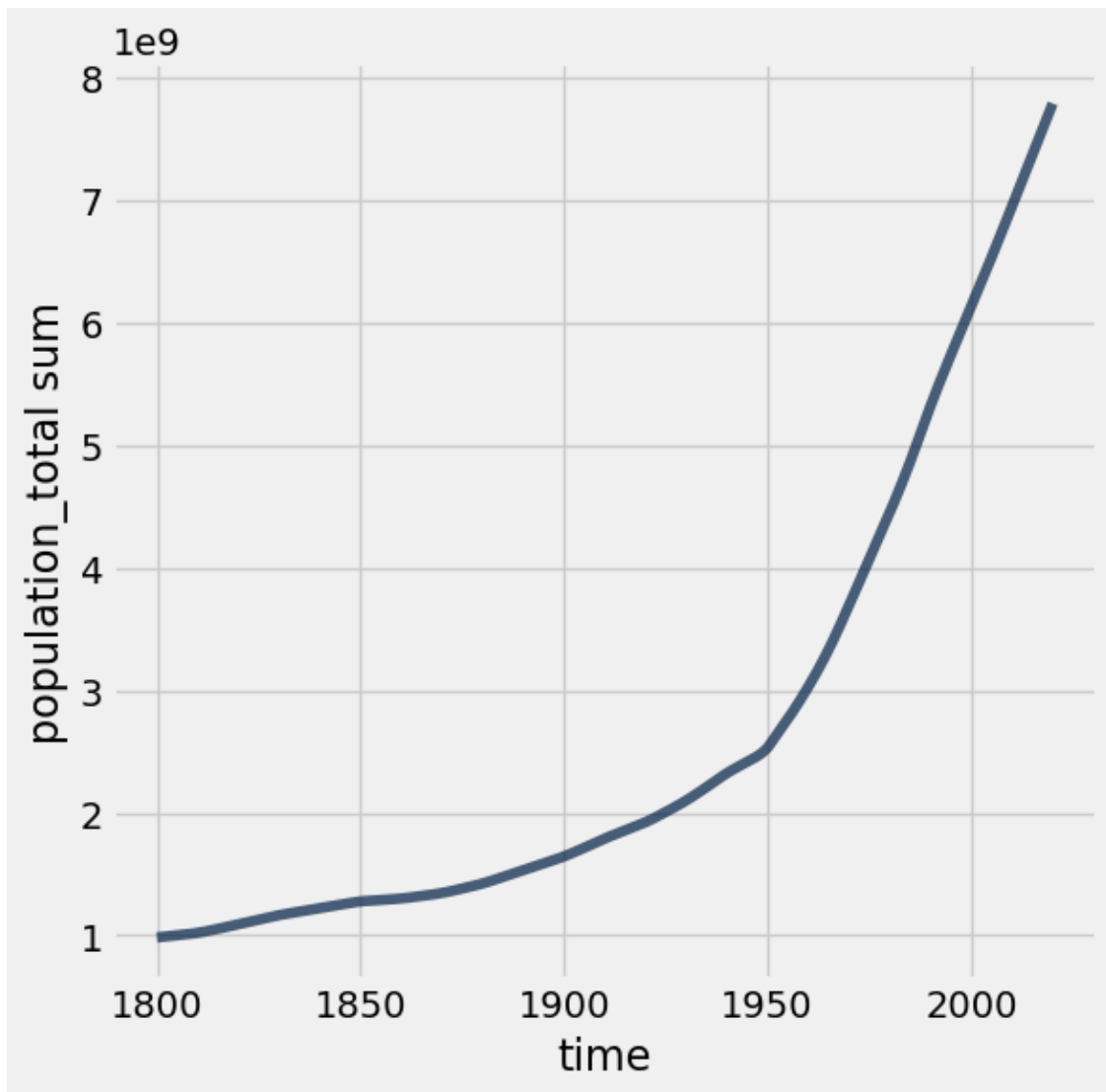
The rate of fertility, or children per woman, in Poland generally decreased between 1950 and 2020, which can be associated with the number of new births in Poland during this time. If everything else is constant, the growth rate of the population would decrease because less children were being born per woman as time passed. Therefore, this trend does help directly explain the decrease in population growth.

Question 8. In one or two sentences, describe the association (if any) that is illustrated by this scatter plot. Does the diagram show any causal relation between between fertility and child mortality?

The y-values of the points that represent the more recent years are lower, meaning that the mortality rate amongst children decreased as time passed. For these points, the x-values are also lower, meaning that the fertility rate amongst women in Poland also decreased. Therefore, the scatter plot suggests that the child mortality rate decreased as the fertility rate decreased, and that there is a causal relationship between fertility and child mortality.

Question 10. Draw a line plot of the **world population** from 1800 through 2020 (inclusive of both endpoints). The world population is the sum of all of the countries' populations. You should use the `population` table defined earlier in the project.

```
In [95]: #Fill in code here
population.where('time', are.between_or_equal_to(1800,2020)).group('time',sum).plot(0,2)
```



Question 6. It is important to study the absolute number of people living in poverty, not just the percent. The absolute number is an important factor in determining the amount of resources needed to support people living in poverty. In the next two questions you will explore this.

In Question 7, you will be asked to write a function called `poverty_timeline` that takes **the name of a country** as its argument (not the Alpha-3 country code). It should draw a line plot of the number of people living in poverty in that country with time on the horizontal axis. The line plot should have a point for each row in the poverty table for that country. To compute the population living in poverty from a poverty percentage, multiply by the population of the country **in that year**.

For this question, write out a generalized process for Question 7. Make sure to answer/include the following:
- What should this function output? - Additionally, **make a numbered list** of the steps you take within the function body. If you added/edited, say, 5 lines in the function body, then it would be good to see the numbers 1 through 5 describing what you did (i.e. what functions/methods you used) in each line and why.

As a tip, after finishing question 7, we recommend polishing up your description of the steps for this question.

The function matches the inputted country name to its corresponding code and ultimately displays a line plot of people in poverty over time.

1: Filter population - Includes only population entries that correspond to the identified country code.

2: Filter poverty - Includes only poverty entries that correspond to the identified country code.

3: Join population and poverty with time - Merges the population and poverty data based on the time entries, in years.

4: Calculate total poverty - Divides the product of the percentage of people in poverty and the total population of the identified country by 100, which outputs the number of people in poverty in the country.

5: Plot - Plot the total number of people in poverty on the y-axis and the years over time, allowing for the display of trends in poverty over time.

6: Labelling x-axis - Years are on the horizontal axis, so the x-axis is labeled as 'Year'.

7: Labelling y-axis - The number of people in poverty is on the vertical axis, so the y-axis is labeled as 'People in Poverty'.

Question 7. Now, we'll actually write the function called `poverty_timeline`. Recall that `poverty_timeline` takes **the name of a country** as its argument (not the Alpha-3 country code). It should draw a line plot of the number of people living in poverty in that country with time on the horizontal axis. The line plot should have a point for each row in the `poverty` table for that country. To compute the population living in poverty from a poverty percentage, multiply by the population of the country **in that year**.

Note: You **should not** return anything from your function. Simply call `plots.show()` at the end of your function body.

Hint 1: This question is long. Feel free to create cells and experiment. You can create cells by going to the toolbar and hitting the + button.

Hint 2: Consider using `join` in your code.

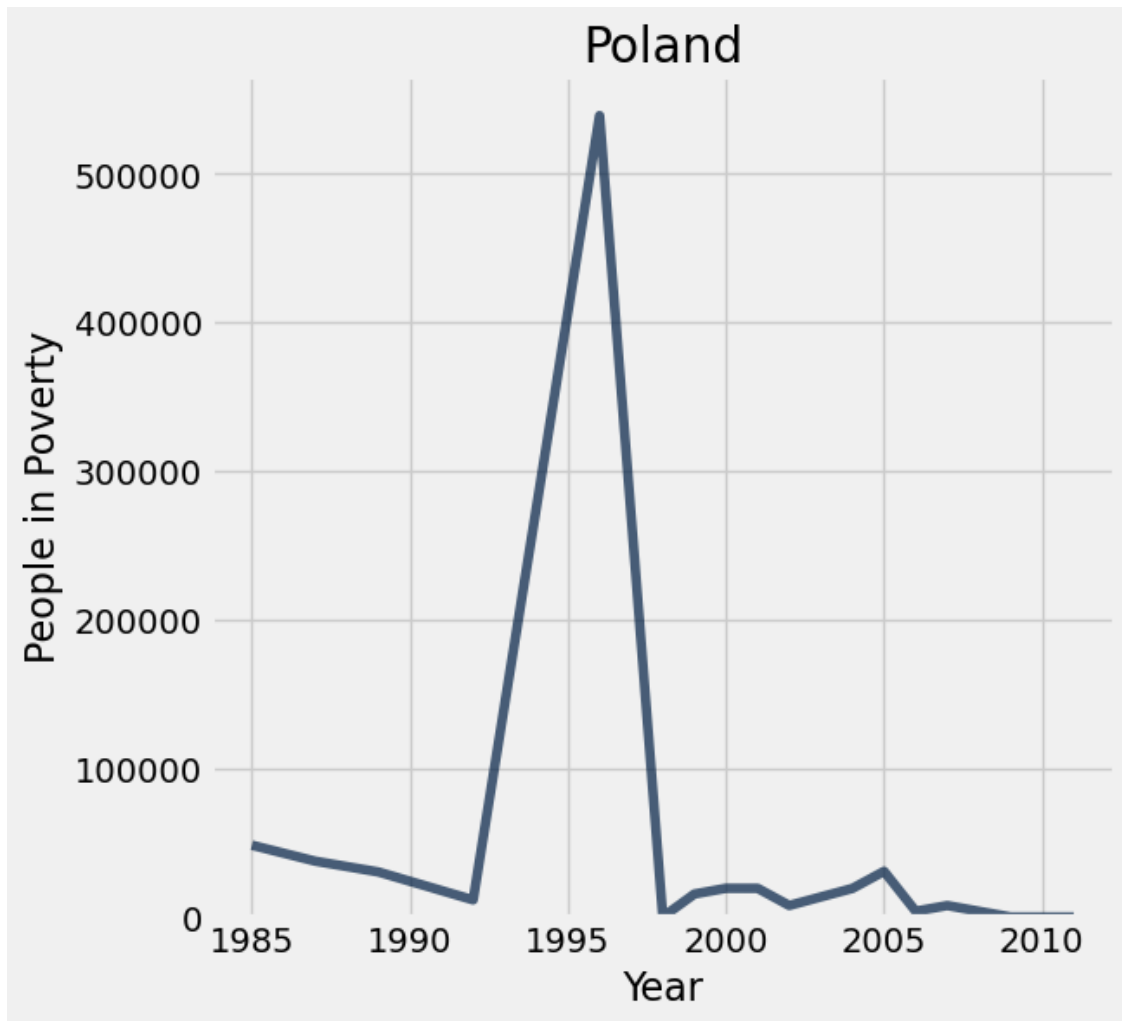
Feel free to use the markdown cell below to plan out your answer, but you needn't fill it in.

Type your answer here, replacing this text.

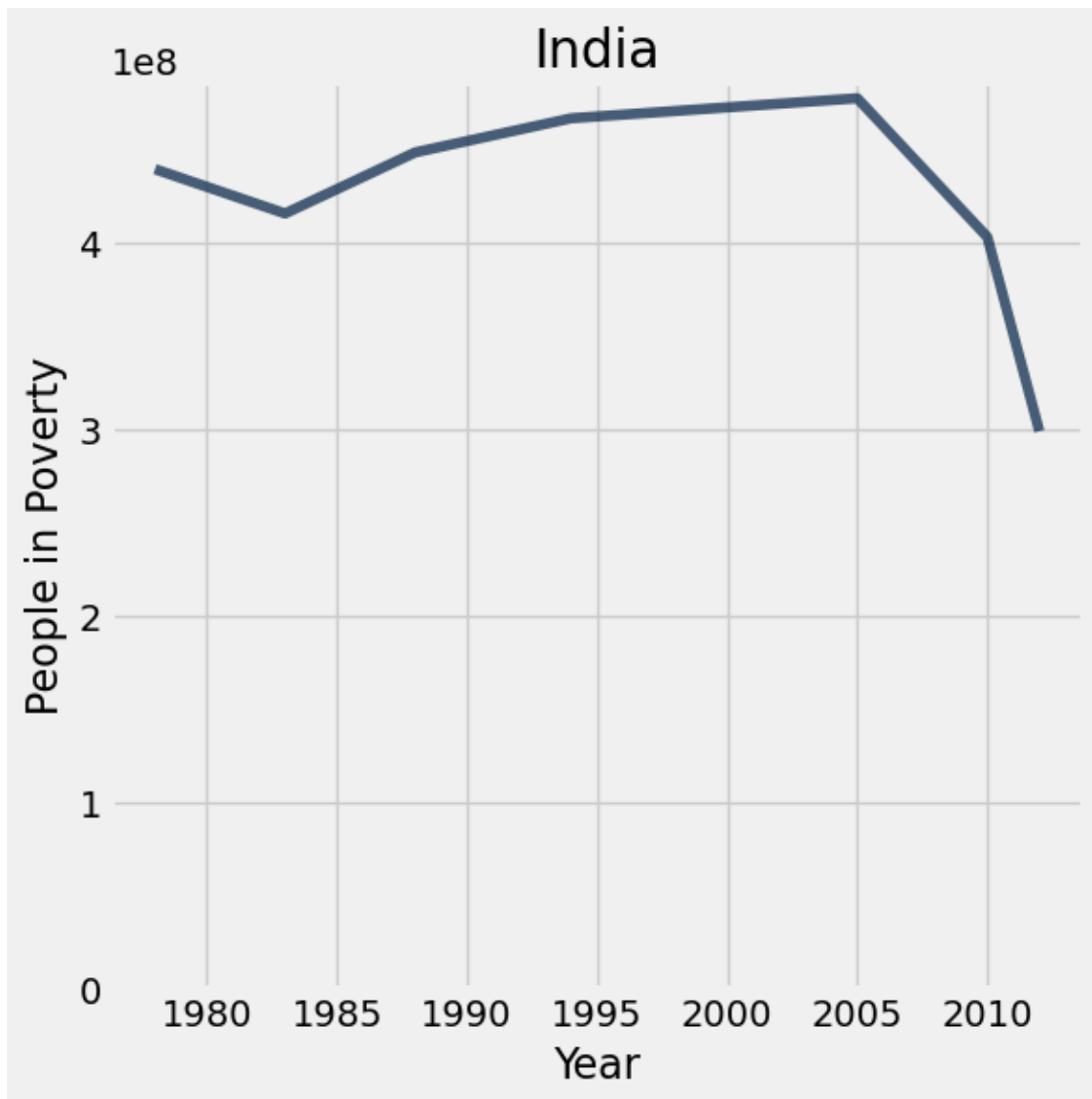
```
In [122]: def poverty_timeline(country):
    '''Draw a timeline of people living in extreme poverty in a country.'''
    geo = countries.select('name', 'country').where('name', are.equal_to(country)).column('country')
    # This solution will take multiple lines of code. Use as many as you need
    pop1 = population.where('geo', are.equal_to(geo))
    pov1 = poverty.where("geo", are.equal_to(geo))
    poppov = pov1.join('time', pop1, 'time').drop('geo_2', 'geo')
    poppov = poppov.with_columns('poverty_total', (poppov.column('extreme_poverty_percent_people') *
    poppov.column('population')).cast('float'))
    poppov.plot('time', 'poverty_total')
    plots.xlabel('Year')
    plots.ylabel('People in Poverty')
    # Don't change anything below this line.
    plots.title(country)
    plots.ylim(bottom=0)
    plots.show() # This should be the last line of your function.
```

```
In [123]: poverty_timeline('Poland')
poverty_timeline('India')
poverty_timeline('Nigeria')
poverty_timeline('China')
poverty_timeline('Colombia')
poverty_timeline('United States')
```

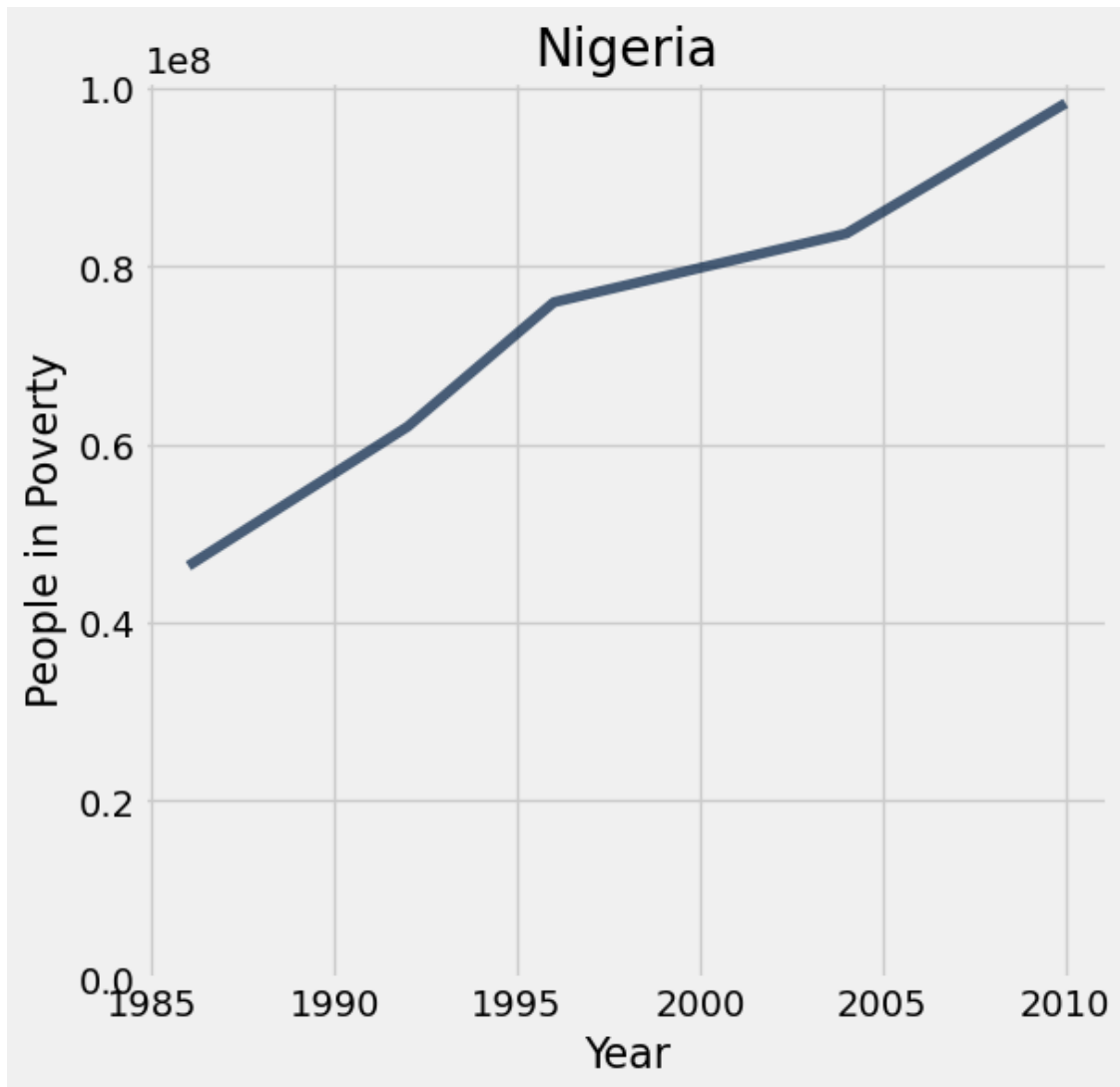
```
/srv/conda/envs/notebook/lib/python3.11/site-packages/datascience/predicates.py:257: SyntaxWarning: Do not use \
warnings.warn("Do not pass an array or list to a predicate. \
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```



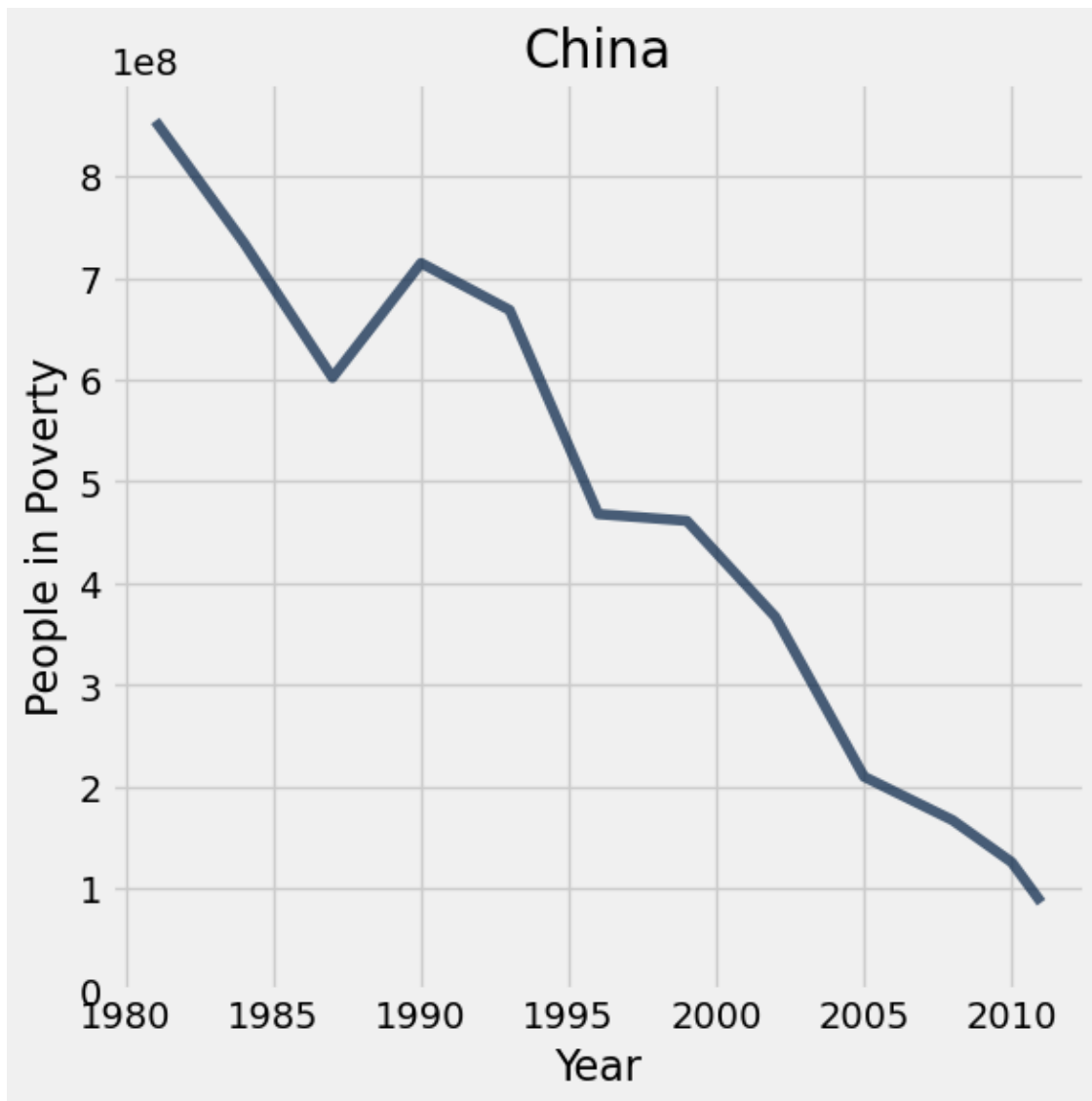
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```



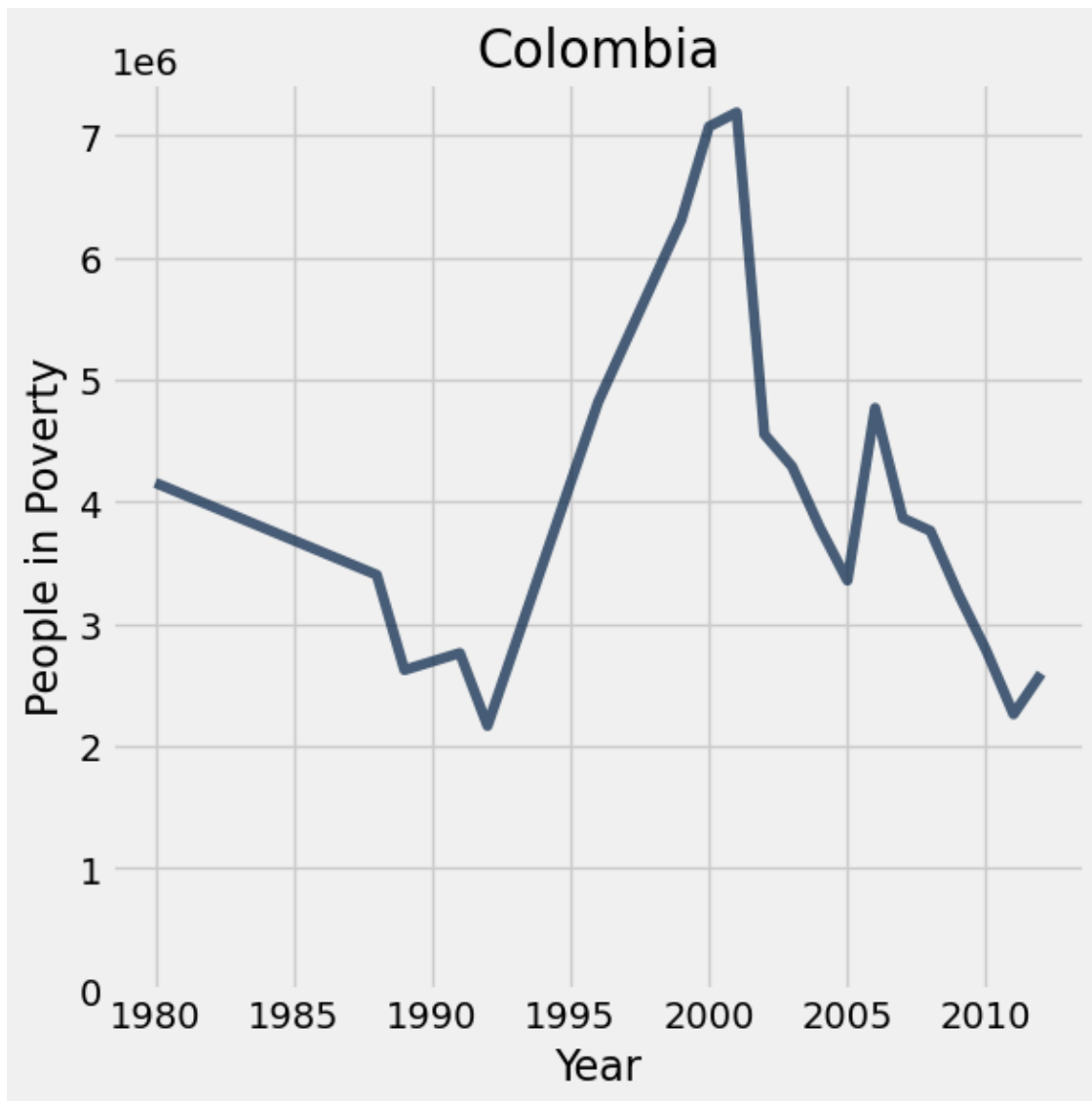
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```
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```
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```



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```

